



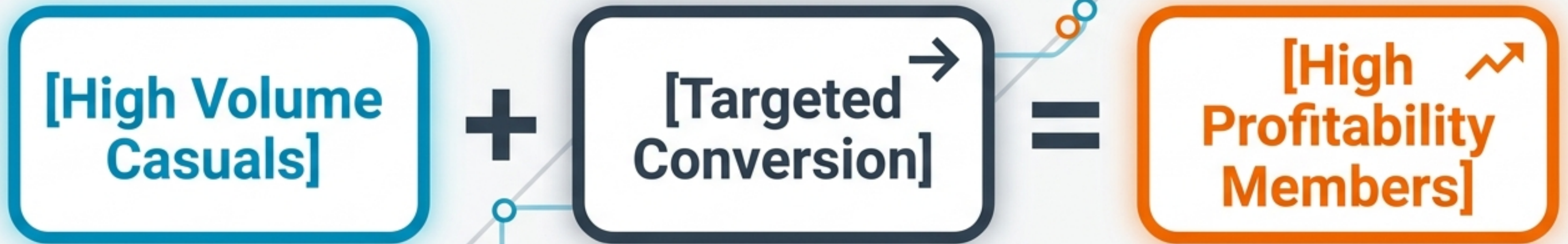
The Cyclistic Divide: Converting Casual Riders to Annual Members

A Data-Driven Marketing
and Growth Strategy



Millions of historical data points synthesized into a definitive roadmap for closing the profitability gap between occasional usage and annual loyalty.

The Business Mandate



The Reality

Annual members are fundamentally more profitable than casual riders, yet casuals represent a massive, untapped volume of usage.

The Mandate

Analyze historical trip data to uncover the exact behavioral divide between these two cohorts. Use these distinct patterns to build a highly targeted marketing bridge that converts single-use riders into recurring revenue.

Methodology & Rigor



Raw Telemetry

Millions of historical ride records ingested (Python, Pandas).



Data Scrubbing

Purging invalid geometries, handling missing values, and removing negative/zero duration errors to ensure absolute fidelity.



Feature Engineering

Extracting behavioral dimensions—Ride Duration, Seasonality, Rush Hour flags, and Weekend indicators.



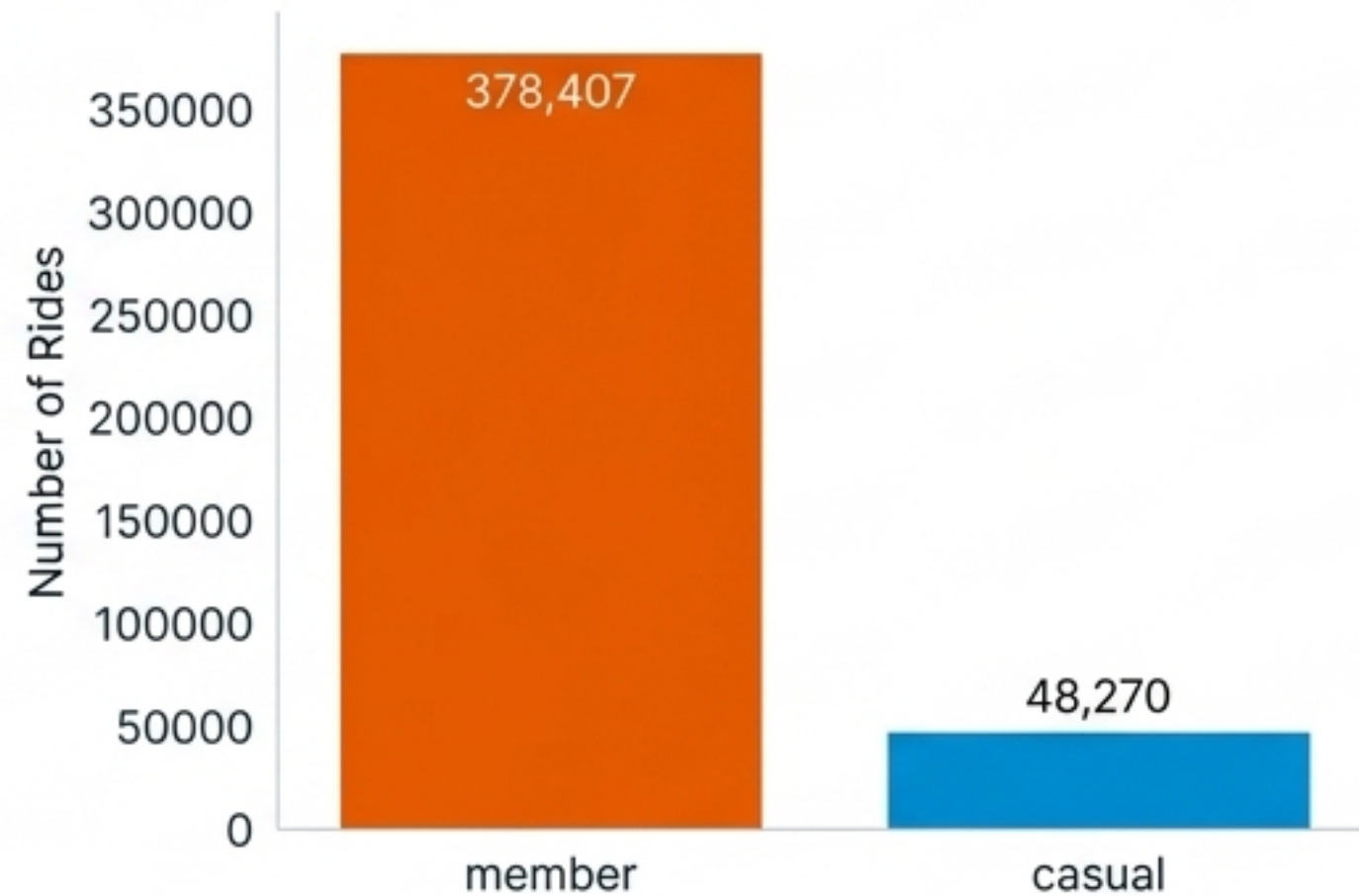
Statistical Validation

EDA reinforced with T-tests and Chi-Square validations.

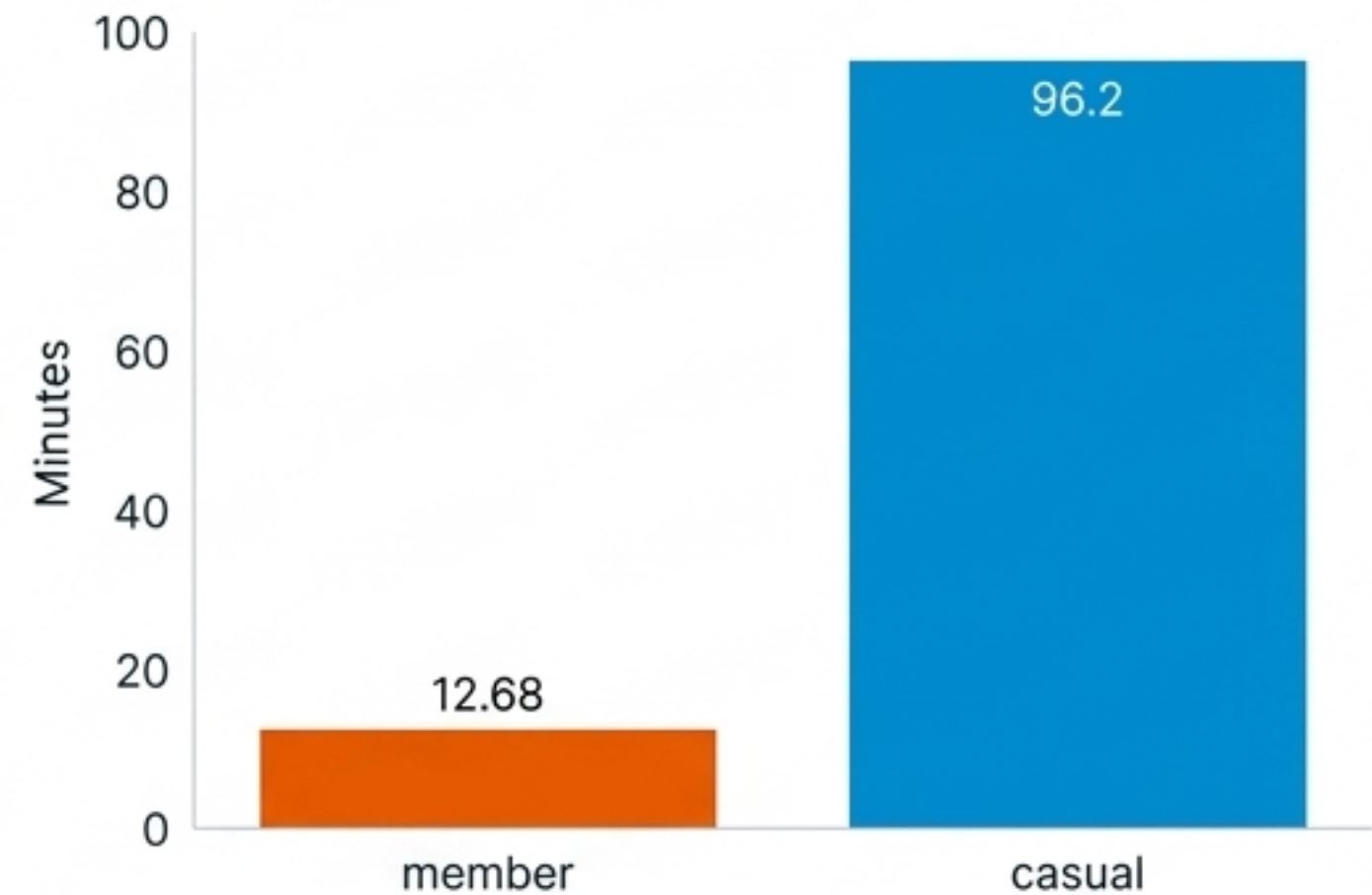
Strategic recommendations built on a foundation of mathematically validated, heavily scrubbed historical telemetry.

The Baseline Split: Volume vs. Duration

The Volume Asymmetry



The Duration Paradox

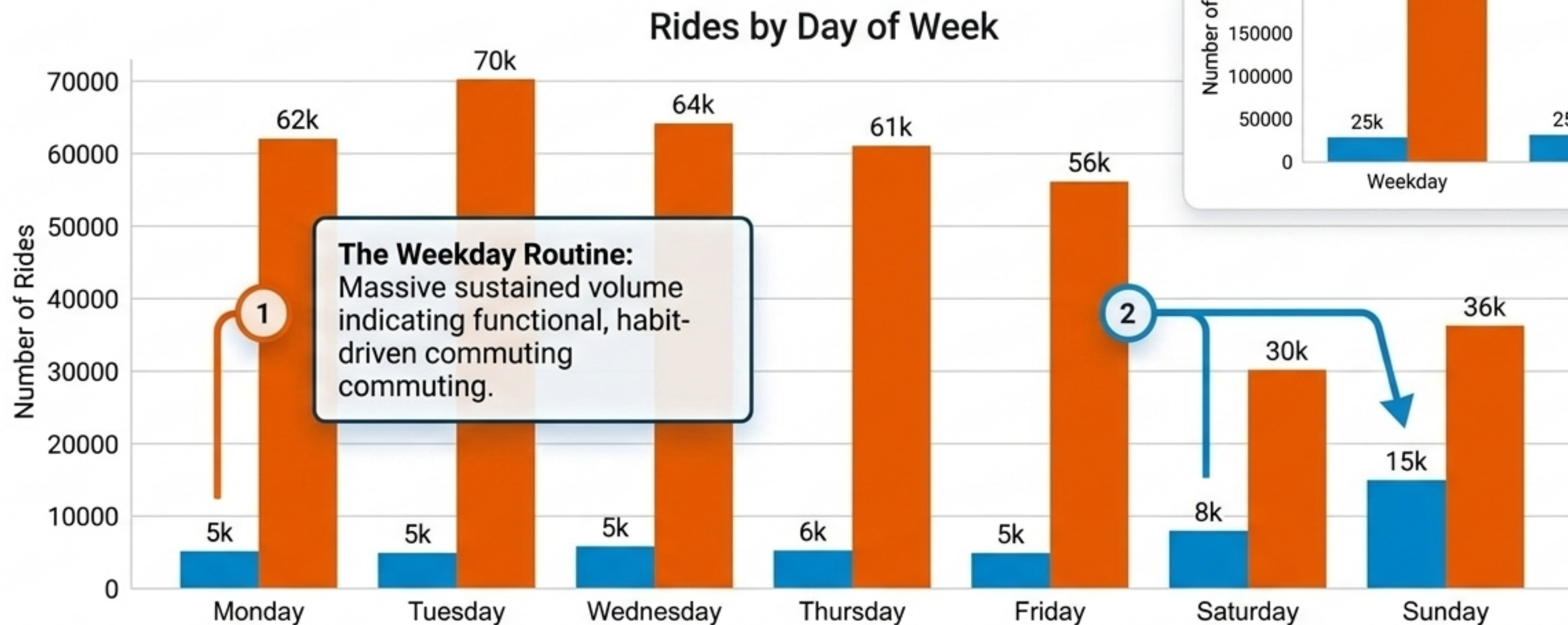


Members use bikes as functional tools for transit. Casuals use bikes as immersive experiences.
To convert casuals, we must bridge the gap between experience and function.

The Rider Persona Matrix

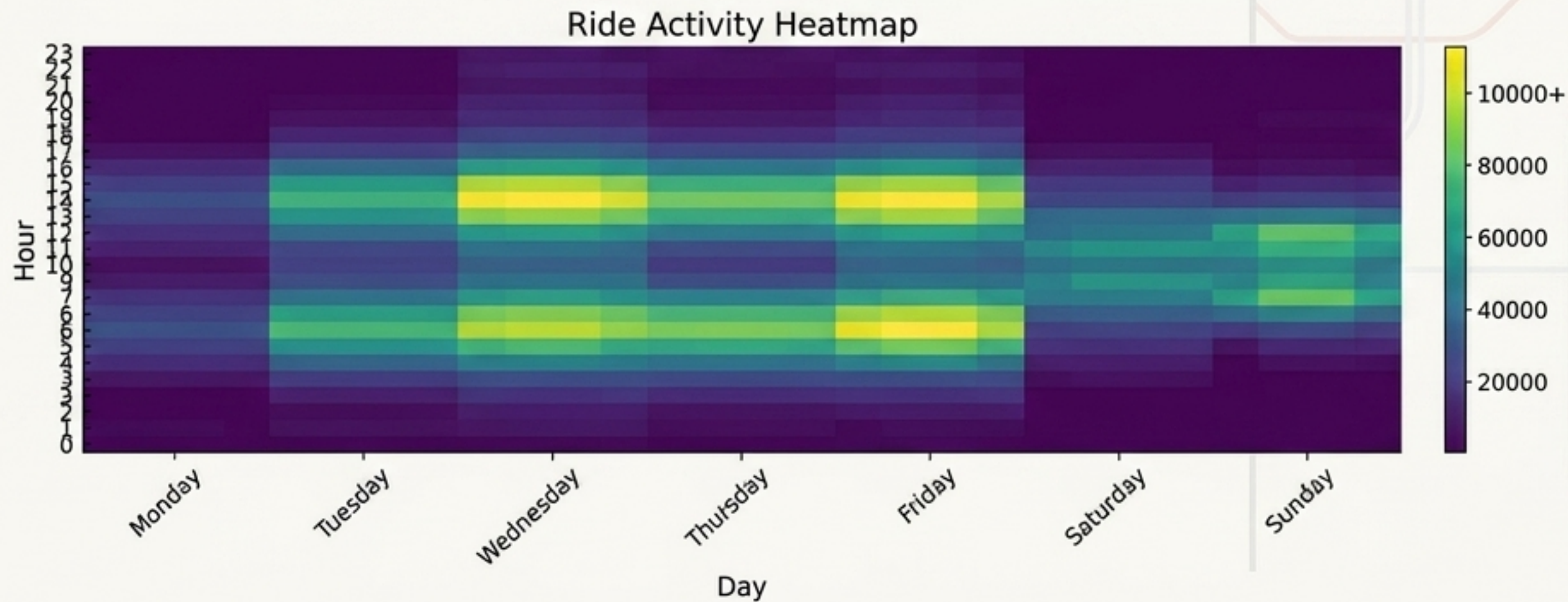
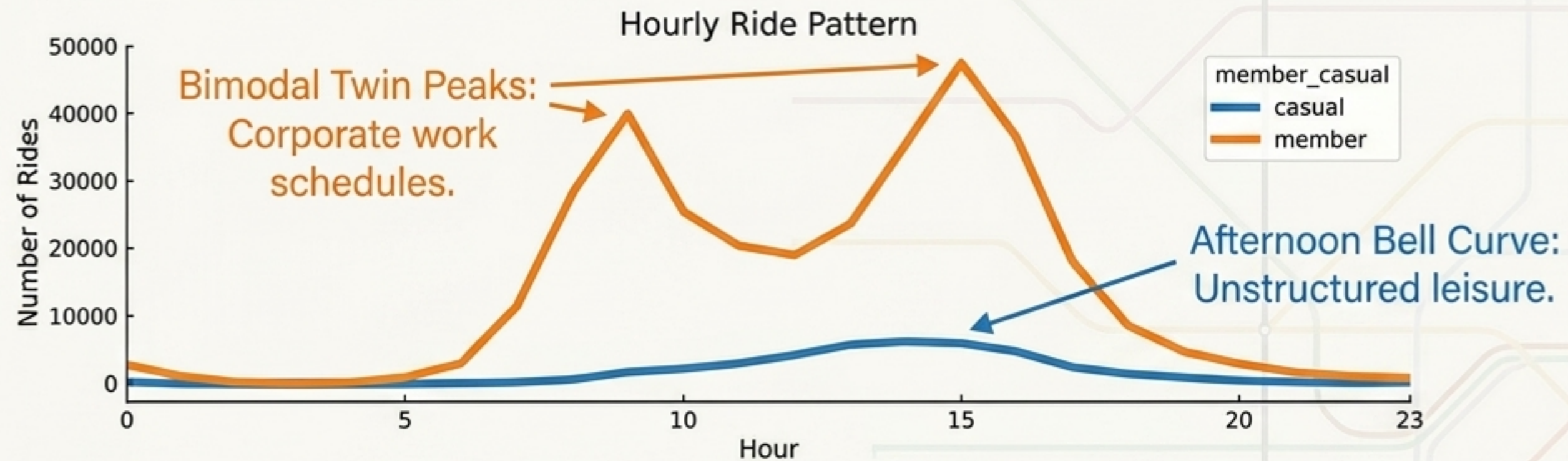
		The Wanderer (Casual)	The Commuter (Member)
	Primary Intent	Leisure & Exploration	A-to-B Transit
	Peak Activity Days	Weekends	Monday-Friday
	Peak Hours	Mid-Afternoon Swell	8 AM & 5 PM Rush
	Average Duration	~96 Minutes	~12 Minutes
	Urban Footprint	Coastal/Tourist Attractions	Commercial/Financial Districts

Behavioral Divergence: The Weekly Cycle



Strategic Takeaway: Casual riding is intensely concentrated on Saturdays and Sundays. Weekend-specific interventions offer the highest probability of **engagement**.

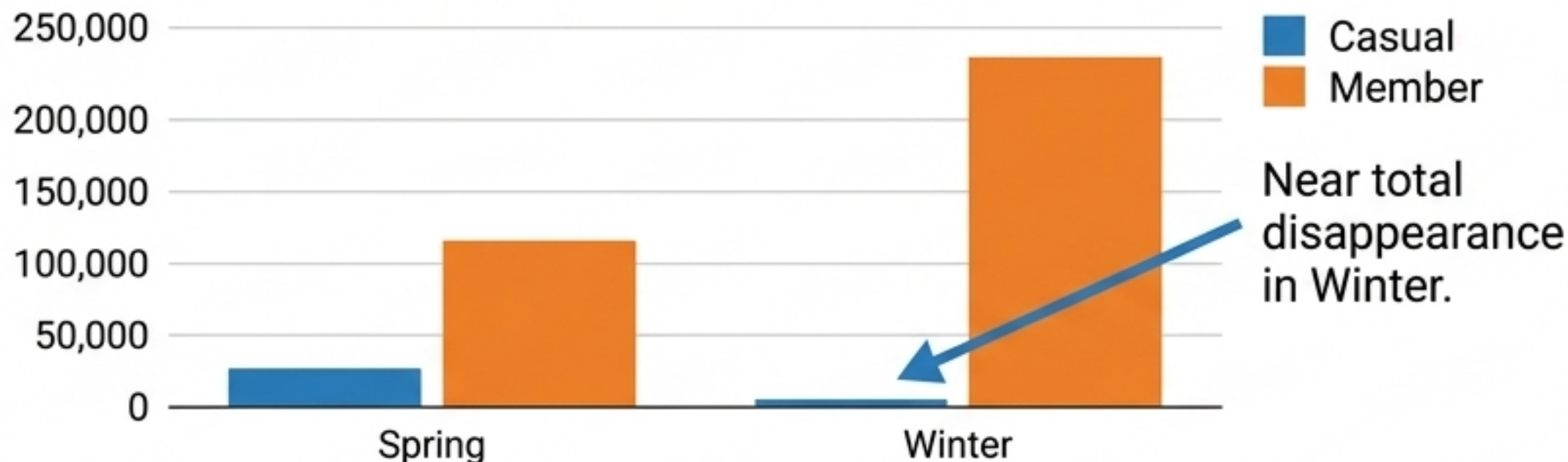
The Daily Urban Rhythm



Member traffic perfectly mirrors corporate work schedules.
Casual traffic aligns with unstructured leisure time.

The Seasonal Swell

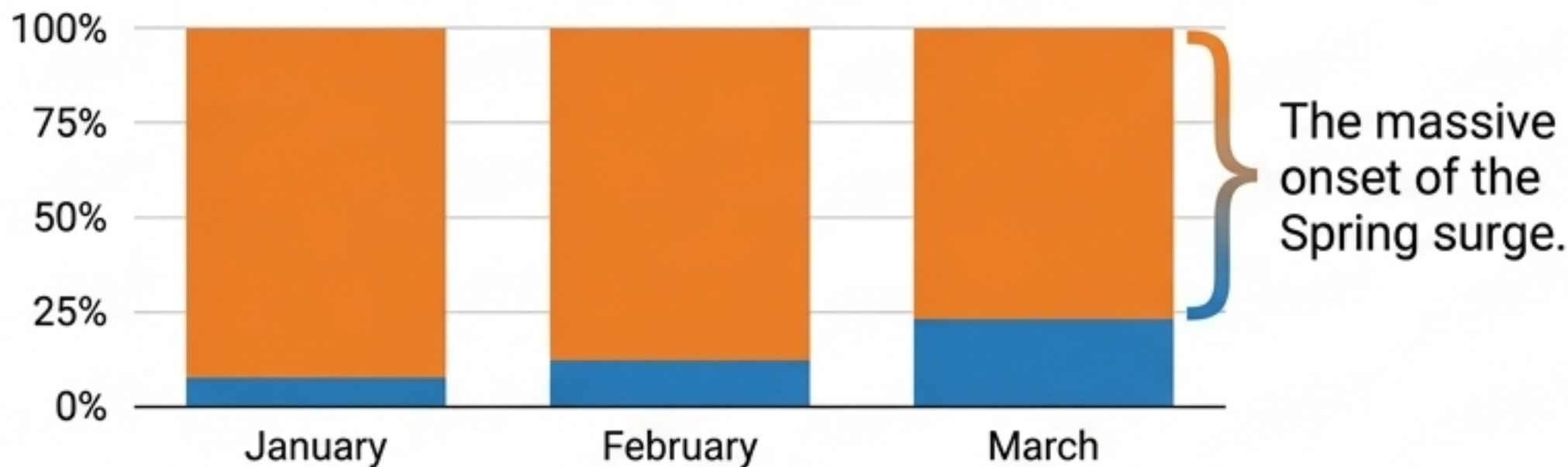
Seasonal Ride Distribution



Baseline Stability vs. Volatility

While Member ridership dips in colder months, the baseline remains incredibly stable compared to the casual cohort.

Monthly Ride Share (%)



Strategic Takeaway: Casual riders are highly elastic to weather. The optimal window for conversion campaigns is the exact onset of the Spring seasonal swell.

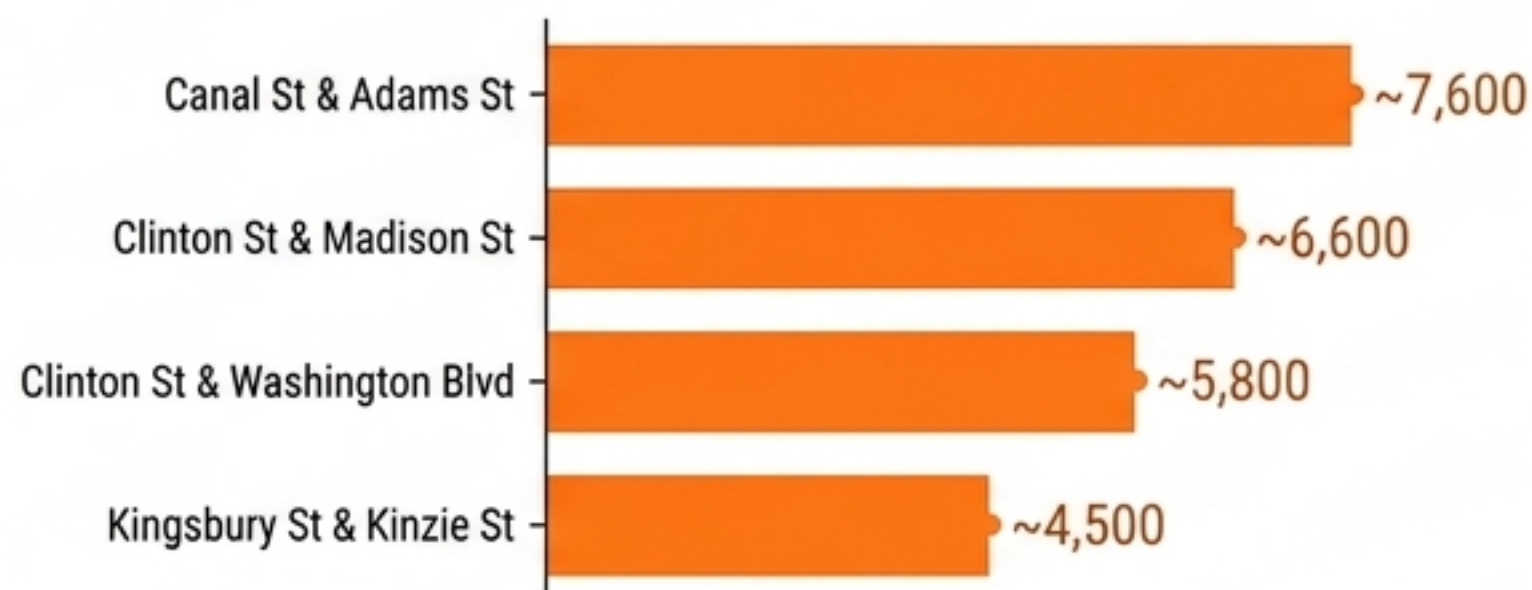
The Geographic Footprint



Financial &
Commercial Districts

The Business Grid

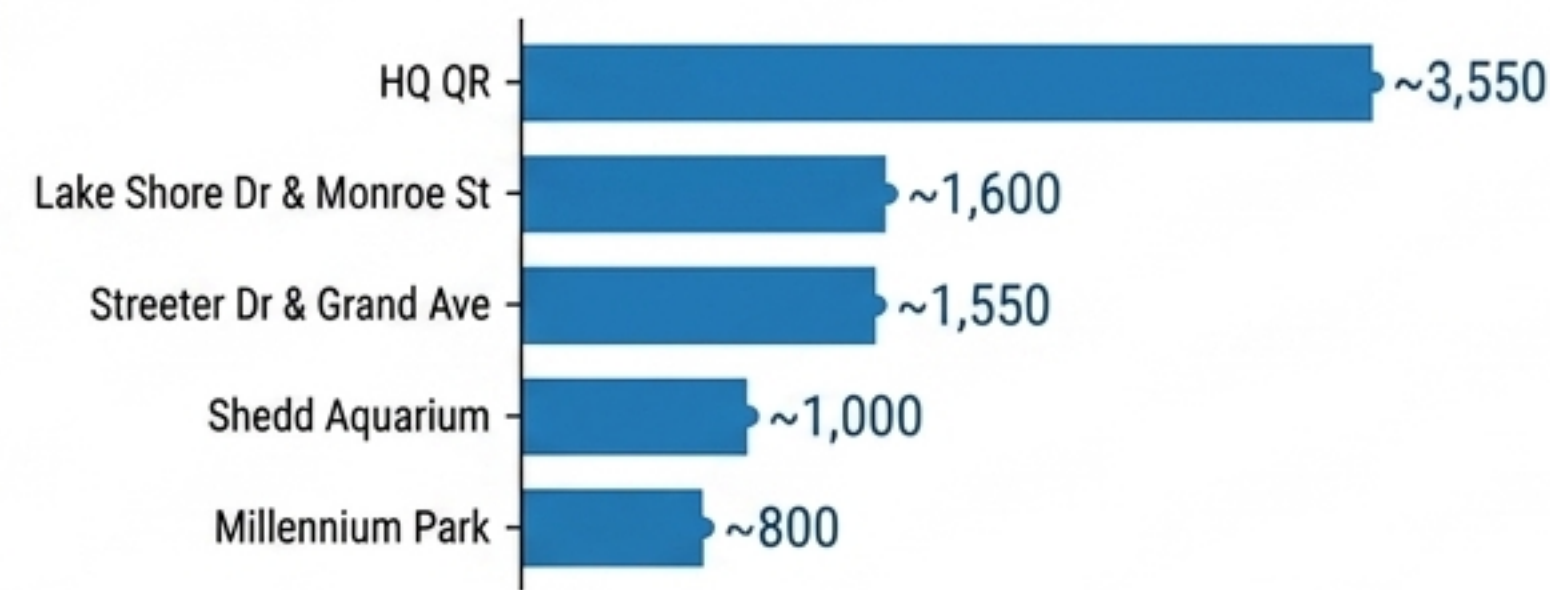
Top 4 Member Start Stations



Tourist Destinations
& Coastal Parks

The Leisure Coast

Top 5 Casual Rider Start Stations

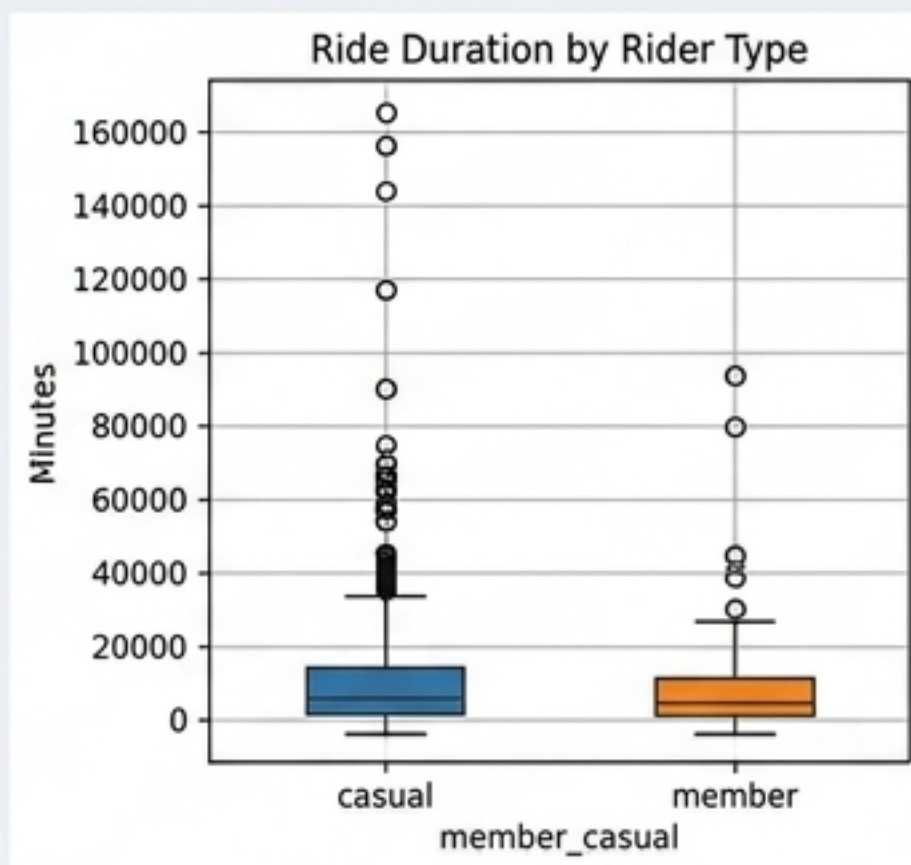


Key Insight: Conversion cannot happen everywhere. Point-of-sale marketing must be hyper-localized to the coastal and tourist hubs where casuals naturally congregate.

Statistical Validation (Proof Behind the Personas)

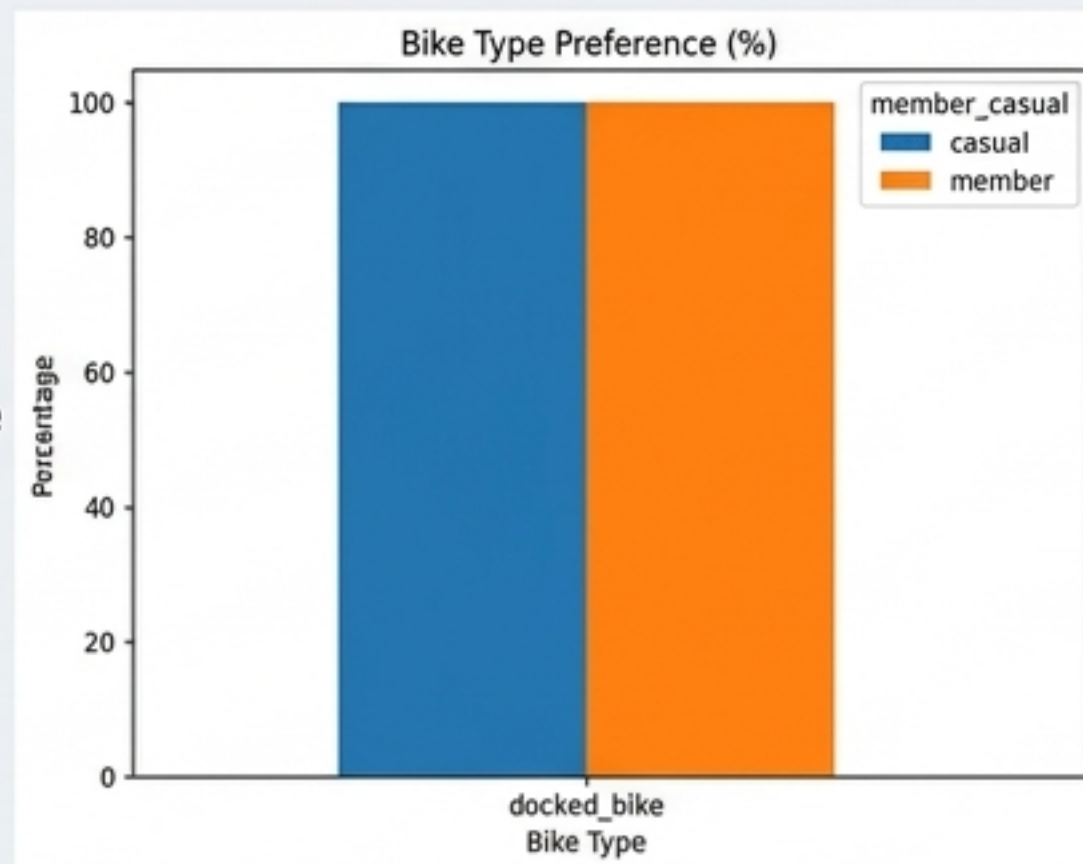
Duration T-Test

The 84-minute gap in average ride duration is highly statistically significant. This represents a fundamental difference in user psychology, not random chance.



Equipment Preference (Chi-Square)

Despite vast behavioral differences, both groups demonstrate an overwhelming shared preference for standard docked bikes.



The data is conclusive. The distinct rider profiles are statistically validated, providing a secure foundation for capital deployment.

The Conversion Blueprint

Three data-backed interventions designed to lower the barrier to annual commitment.

Leverage Time



Aligning campaigns with the established weekend and summer swells.

Leverage Location



Intercepting casuals at their specific geographic origins.

Leverage Value



Structuring pricing to turn long, single-use rides into habitual usage.

Strategy 1: Targeted Time-Based Campaigns

January							February							March						
S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S
					1	2			1	2	3	4	5			1	2	3	4	
3	4	5	6	7	8	9	6	7	8	9	10	11	12	5	6	7	8	9	10	11
10	11	12	13	14	15	16	13	14	15	16	17	18	19	12	13	14	15	16	17	18
17	18	19	20	21	22	23	20	21	22	23	24	25	26	19	20	21	22	23	24	25
24	25	26	27	28	29	30	27	28						26	27	28	29	30	31	
31																				
April							May							June						
S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S
						1	1	2	3	4	5	6	7			1	2	3	4	
2	3	4	5	6	7	8	8	9	10	11	12	13	14	5	6	7	8	9	10	11
9	10	11	12	13	14	15	15	16	17	18	19	20	21	12	13	14	15	16	17	18
16	17	18	19	20	21	22	22	23	24	25	26	27	28	19	20	21	22	23	24	25
23	24	25	26	27	28	29	29	30	31					26	27	28	29	30		
30																				
October							November							December						
S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S
					1	2			1	2	3	4	5			1	2	3	4	
3	4	5	6	7	8	9	7	8	9	10	11	12	13	5	6	7	8	9	10	11
10	11	12	13	14	15	16	14	15	16	17	18	19	20	12	13	14	15	16	17	18
17	18	19	20	21	22	23	21	22	23	24	25	26	27	19	20	21	22	23	24	25
24	25	26	27	28	29	30	28	29	30					26	27	28	29	30	31	
31																				



Recommendation

Target casual riders with Weekend-Only or "Summer Pass" micro-memberships as a stepping stone to full annual commitment.



Data Backing

Directly addresses the intense Saturday/Sunday volume spikes and the massive Spring/Summer seasonal surge identified in the temporal analysis.



Execution

Launch aggressive digital push notifications on Thursday evenings targeting previous casual users, offering instant weekend unlocks.

Strategy 2: Geographic & Point-of-Sale Nudges



Top 10 Start Stations



Recommendation

Deploy hyper-local OOH advertising and instantaneous QR-code trial upgrades exclusively at top leisure stations.

Data Backing

Focus capital solely on the top 10 tourist hubs (HQ QR, Shedd Aquarium, Millennium Park) where casual durations originate.

Execution

Intercept the user at the exact moment of high-duration leisure intent.

Strategy 3: Behavioral Incentives



Recommendation

Introduce a one-month discounted trial and a duration-based loyalty reward program.

Data Backing

Casual riders already demonstrate high-duration usage. A loyalty program rewards this specific behavior, making an annual membership mathematically logical.

Execution

Automated email triggers to any casual rider who logs a ride over 60 minutes, demonstrating exactly how much they would have saved as a member.

Executive Dashboard & Next Steps

Total Rides Analyzed

426k+

Leisure usage data

Member / Casual Split



Avg Duration Delta

+84 mins

Higher casual rider duration

Conversion Blueprint: Finalized Next Steps



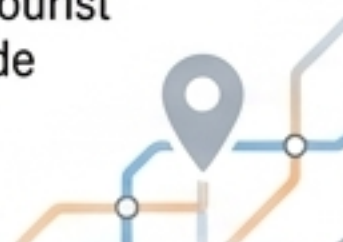
TIME: Deploy Spring & Weekend Passes

Deploy Spring & Weekend Passes to spring our calendar and clock.



LOCATION: Activate QR-Trials at Tourist Hubs

Activate QR-Trials at Tourist Hubs, in easy to provide contrirntos location.



VALUE: Trigger Duration-Based Loyalty Rewards

Trigger duration-based charity to loyalty rewards.



The analytics workflow is complete. Cyclistic now possesses the behavioral profiles, the geographic targets, and the exact temporal windows required to deploy marketing capital and maximize ROI.